

MOHD SAHIL ANSARI

DATA SCIENCE | MACHINE LEARNING | FRAUD & RISK ANALYTICS

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PROFESSIONAL SUMMARY

Data Science and Machine Learning professional with a B.Sc. in Computer Science and hands-on financial-services experience spanning fraud investigation, AML/KYC, transaction risk, and dispute analysis. Combines practical financial-crime domain knowledge with applied machine learning skills in Python, SQL, Scikit-learn, feature engineering, model evaluation, NLP, explainable AI, and Flask deployment. Experienced in translating transaction and customer-risk signals into investigation-oriented decisions and building end-to-end ML solutions.

CORE TECHNICAL & DOMAIN SKILLS

Languages & Data	Python, SQL, MySQL, Pandas, NumPy
Machine Learning	Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation, Ensemble Learning
Algorithms	Random Forest, Logistic Regression, Decision Tree, KNN, SVM, Naive Bayes, Gradient Boosting, XGBoost, AdaBoost, Extra Trees, Ridge, Lasso
NLP	Text Preprocessing, Tokenization, Stop-word Removal, Stemming, TF-IDF, Bag-of-Words, CountVectorizer
Explainability & Risk	SHAP, Transaction Risk Scoring, Fraud Analytics, Imbalanced Classification, Leakage-aware Feature Engineering
Visualization & Tools	Matplotlib, Seaborn, WordCloud, Jupyter, Google Colab, VS Code, Git, GitHub
Deployment	Flask, Joblib/Pickle, API integration
Financial Crime Domain	Fraud Investigation, AML, KYC, PEP & Sanctions Screening, Transaction Monitoring, Dispute Investigation, Chargebacks

PROFESSIONAL EXPERIENCE

Fraud & Dispute Investigation Analyst | Teleperformance

Jul 2025 – Present

- Investigate card-fraud and transaction-dispute cases by analyzing transaction activity, customer information, and supporting evidence to identify potentially unauthorized or suspicious activity.
- Perform structured case investigations and evidence assessment, applying risk-based judgment to determine appropriate case outcomes while maintaining investigation quality and documentation standards.
- Initiate VISA chargeback processes for eligible disputes following investigation and validation of transaction and case evidence.
- Work within financial-services processes where accuracy, evidence quality, risk assessment, and timely case resolution are critical.

AML/KYC Analyst | Firstsource

13 months

- Performed AML/KYC investigations covering customer due diligence, identity documentation, transaction and activity indicators, PEP screening, sanctions screening, and risk assessment.
- Reviewed real-time alerts and customer information to support risk-based account-opening and investigation decisions.
- Analyzed customer and transaction information to identify risk indicators and escalate cases requiring additional review.
- Managed high-volume casework while maintaining investigation accuracy, documentation quality, and adherence to compliance procedures.

SELECTED PROJECTS

Fraud Detection & Transaction Risk Scoring System | Python, Pandas, Scikit-learn, SHAP, Flask, PAYSim

- Built an end-to-end transaction fraud detection and investigation workflow using the synthetic PAYSim financial transaction dataset.
- Designed chronological train/validation/test splits to reduce temporal leakage and engineered leakage-safe historical sender/receiver behavioural features using only prior transaction activity.
- Created transaction, balance, behavioural, temporal, and categorical features; addressed severe class imbalance through training-data balancing while preserving the original class distribution for validation and testing.
- Trained a Random Forest classifier and evaluated performance using precision, recall, F1-score, PR-AUC, ROC-AUC, and confusion matrix; generated transaction-level fraud risk scores.
- Integrated SHAP TreeExplainer for model interpretability and built evidence-based investigation guidance, human-in-the-loop decisions, and a Flask investigation interface.
- On the synthetic PAYSim test set, the final model achieved 1.00 ROC-AUC, 0.9996 F1, and 0.999996 PR-AUC; results are dataset-specific and not representative of real-world bank fraud performance.

Movie Recommendation System | Python, Pandas, Scikit-learn, NLTK, Flask, TMDB API

- Built a content-based movie recommendation engine using TMDB 5000 Movies and Credits data.
- Combined movie overview, genres, keywords, cast, and crew into text features; applied preprocessing and Porter stemming before vectorization.
- Used CountVectorizer and cosine similarity to generate recommendations and deployed the workflow through Flask with TMDB API integration for movie details and posters.

SMS Spam Classifier | Python, Pandas, Scikit-learn, NLTK

- Developed an NLP pipeline for SMS spam classification using text cleaning, normalization, stemming, TF-IDF, and supervised learning.
- Compared multiple classification and ensemble approaches using precision, recall, F1-score, confusion matrix, and classification reports, with focus on performance beyond accuracy.
- Applied class-imbalance handling during model development and selected models based on classification effectiveness.

EDUCATION

B.Sc. Computer Science | University of Mumbai | 2023

PROFESSIONAL LINKS

GitHub: github.com/mohd2548 | Portfolio: <https://mohd2548.github.io/portfolio-site/>